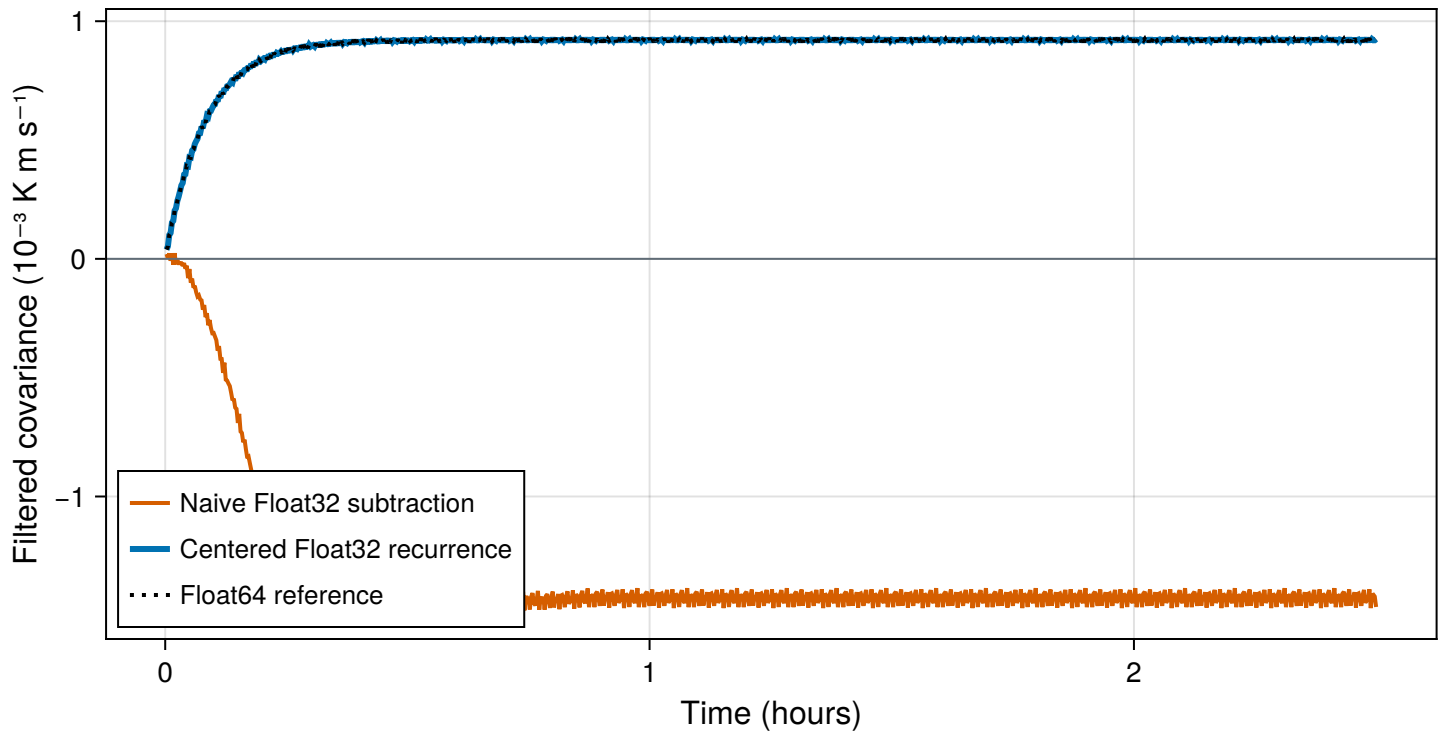


A small covariance can get the wrong sign in Float32

Synthetic arithmetic test: 90,000 samples, $\Delta t = 0.1$ s, filter time = 300 s.
Identical Float32 input samples are used in the Float64 reference.



Final covariance: naive -1.465 | centered 0.925 | reference 0.924

Units: 10^{-3} K m s⁻¹. Centered recurrence relative error: 0.020%.

Why this matters for the closure

Subtracting two much larger filtered products loses the small turbulent flux. Here it reverses its sign. A centered covariance update retains the signal that controls the added diffusivity.

Implementation checks support the arithmetic result

119 CPU feature checks cover filtering, flux signs and guards, support, implicit conservation, and checkpoint continuation. Separate CPU runner and export checks cover shapes, native heights, schedules and provenance.

Boundary of the evidence: synthetic arithmetic and CPU checks are not evidence of improved LES fidelity. GPU validation and matched LES remain pending. Reproducer, exact plotted CSV, and source hashes accompany this section.